# SkyTrack Project

## Completion Probability, Cost & Time Trade-offs, and Scenario Analysis

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# EXECUTIVE SUMMARY

AeroDynamics Inc. is preparing to launch SkyTrack, a state-of-the-art drone designed to transform supply chain operations. As a first mover in an emerging market, the company has prioritized speed to market, setting an ambitious 90-day launch deadline. Meeting this target requires careful coordination of interdependent tasks under strict resource and time constraints. This report applies Program Evaluation and Review Technique (PERT), Critical Path Method (CPM), and project crashing to evaluate the baseline plan, identify risks, and explore strategies for balancing time, cost, and performance.

PERT provides a probabilistic approach to scheduling by using three time estimates for each task to model uncertainty, while crashing shortens project duration by allocating additional resources to critical path activities. Together, they allow managers to plan realistically and intervene when deadlines are threatened. Historical cases such as the U.S. Navy's Polaris program and the 2004 Athens Olympics highlight their relevance in complex, time-sensitive projects.

The SkyTrack baseline plan consisted of 15 tasks across two parallel critical paths, with an expected completion time of 103 days. PERT analysis showed less than a one percent probability of finishing in 90 days, making the original schedule unrealistic. To reach a 95 percent probability, the duration must be reduced to 81 days, requiring aggressive crashing of seven activities. This reduces indirect costs but adds $18,500 in direct costs and places significant strain on resources.

Cost–time trade-off analysis identified 91 days as the optimum schedule, minimizing total project cost at $131,200 compared with $135,500 for the baseline. At this point, the probability of finishing within 90 days rises to 42 percent, offering a balance between efficiency and risk. The shortest possible schedule of 79 days achieves a 98 percent probability but raises costs to $138,900, showing diminishing returns from excessive crashing.

What-if analysis highlighted the project's sensitivity to external risks. A supplier delay scenario extended the duration to 112 days, with full crashing still only achieving a 64 percent chance of on-time delivery. In contrast, reducing the duration of Component Specification Outlining lowered the baseline to 98 days and shifted the cost-optimal point to 89 days at $128,900, improving both cost and schedule feasibility.

Overall, the analysis confirms that the 90-day target cannot be met under the baseline plan. Success will depend on combining selective crashing with strong supplier management and continuous monitoring of the critical path. If cost efficiency is prioritized, the 91-day schedule is recommended. As deadline certainty is essential here, the 81-day aggressive crash plan should be adopted, supported by strict governance and contingency planning.

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# A. INTRODUCTION

AeroDynamics Inc., a technology company specializing in drone innovation, is planning to launch a state-of-the-art drone named SkyTrack that is designed to reform the supply chain and business operations. As a groundbreaking innovation in a newly emerging industry, SkyTrack offers AeroDynamics the opportunity to establish its position in the drone market. AeroDynamics aims to capitalise on the momentum of early adoption by introducing the product as soon as possible, making speed to market a decisive factor for success.

To achieve the target, AeroDynamics has set an ambitious 90-day deadline for the product launch. Rebecca Sinclair (AeroDynamic's Director of Innovation) has identified three main goals to ensure SkyTrack's successful launch: 1) Resilient supply chain strategy for smooth sourcing and production, 2) Effective inventory management for adequate stock and projected demand, and 3) Reliable logistics system for product delivery. These objectives must be pursued within the project's priority matrix, illustrated in Table 1 below, where time is the critical constraint, cost in the area for enhancement, and scope as the most flexible actor.

**Table 1**
*SkyTrack's Priority Matrix*

| | Cost | Time | Scope |
|---|---|---|---|
| Constraint | | × | |
| Enhance | | | × |
| Accept | × | | |

In meeting these objectives, the SkyTrack team would face and solve challenges, such as market demand fluctuation, potential supplier side disruptions, and the need to balance trade-offs between time, cost, and scope. Although SkyTrack's team has already established a solid project plan, evaluating the initial design is essential to identify its weak points and risks. This evaluation will provide a complete picture of the project under the expected, worst-case, and most optimal conditions.

Thus, this report analyzes the current plan and condition of the project using project management methods such as Program Evaluation Review Technique (PERT), Project Crashing, and What-if analysis. It provides answers to five key scenarios designed to identify potential scheduling decision and their cost implications, through the following:

1. The probability of completing the project in 90 days (in line with the target)
2. The tasks and costs required to achieve a 95% likelihood of completion within the targeted timeframe
3. The optimum cost-time point of the project duration

4. The minimum expected completion time and cost
5. The impact of supplier partnerships and reduced task durations on the project schedule

The report also explains the tools' relevance and breaks down the results. It concludes with recommendations for AeroDynamics to strengthen SkyTrack project management and ensure a successful launch.

# B. LITERATURE REVIEW

Project management involves planning, organizing, and controlling resources to achieve objectives within scope, time, and cost constraints. Managers often face tight schedules and limited budgets, making the balance between cost and time critical. Cost and time analysis provides a structured way to decide how project durations can be shortened without reducing scope or quality. In this context, two widely used techniques, the Program Evaluation and Review Technique (PERT) and project crashing, help improve planning accuracy and support schedule compression when deadlines are threatened.

## 1. Program Evaluation and Review Technique

The Program Evaluation and Review Technique (PERT) is a statistical project management technique used to evaluate the tasks required to finish a project, focusing on the time each task may take and determining the shortest possible overall project duration. Unlike rigid schedules, it accounts for uncertainty by allowing planning even when task details and durations are not precisely known. In solving a problem with PERT, the project is divided into individual activities, the necessary preceding tasks for each activity are identified, and time estimates are assigned to their completion. Once this information is established, the rest of the analysis follows a straightforward, mechanical process (Clark, 1962).

PERT is particularly useful in research and development, new product launches, or innovative technology deployments (Meegle, 2025). Developed in 1959 by the U.S. Navy's Special Projects Office for the Polaris Fleet Ballistic Missile program, PERT addressed the challenge of coordinating thousands of unprecedented tasks, many extending years into the future. Because no historical data existed to support reliable estimates, PERT was designed as a statistical tool to analyze task durations and quantify schedule variability. Its successful application helped the Polaris program finish ahead of schedule, cementing PERT as a foundational project management methodology.

PERT acknowledges three key elements that shape the success of research and development projects: time, resources, and technical performance requirements. It treats time as the central variable that captures how resources are allocated and how performance targets are met. By using time as the common measure, PERT translates uncertainties into quantifiable terms, making it possible to plan and manage projects that push the boundaries of current knowledge, where little prior experience exists (Stauber, Douty, Fazar, Jordan, Weinfeld, & Manvel, 1959).

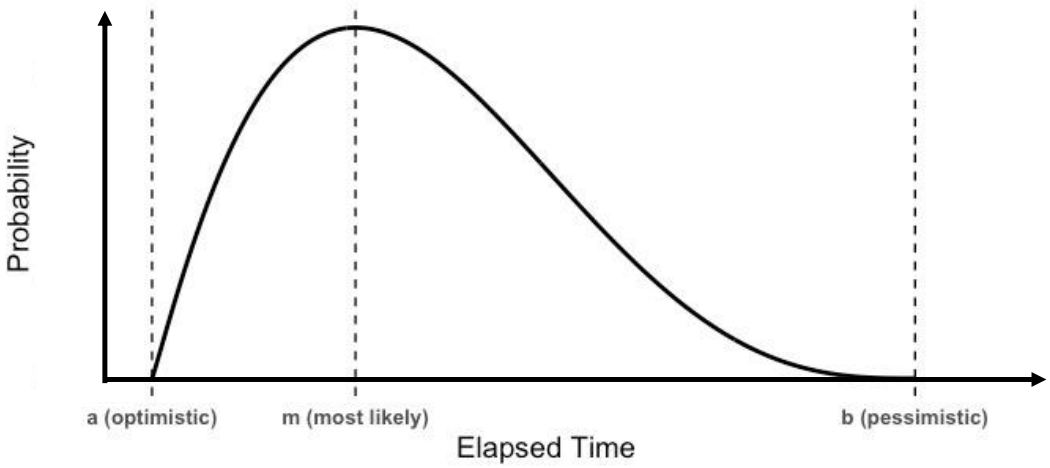
The strength of PERT lies in its probabilistic approach. Rather than relying on a single estimate of the task duration, PERT employs a three-point system: the optimistic time estimate

($a$), the most likely or normal time estimate ($m$), and the pessimistic time estimate ($b$). Each time estimate should be provided by experts familiar with how the activity is carried out, rather than relying solely on management's estimates that were previously used to create the schedule (Clark, 1962).

The Beta distribution (Figure 1) was chosen for PERT because it provides a realistic and practical way to represent uncertain activity times. Unlike the normal distribution, it is bounded between the optimistic (a) and pessimistic (b) estimates and can take on different shapes (symmetric or skewed) to reflect real-world uncertainty. This flexibility makes it suitable for modeling tasks that are more likely to face delays than early finishes (Clark, 1962).

**Figure 1**

*Estimating the Elapsed-time Distribution*




By adopting the Beta distribution, PERT can use simple formulas for the expected time estimate $t_e$ and the variance of the distribution $\sigma^2(t_e)$ :

$$t_e = \frac{(a + 4m + b)}{6} \tag{1}$$

$$\sigma^2(t_e) = \left(\frac{b - a}{6}\right)^2 \tag{2}$$

Once expected times are determined, they are incorporated into a project diagram to identify the critical path, which is the longest chain of dependent activities that defines the minimum project duration. At the same time, the analysis highlights which tasks have slack, indicating the degree of scheduling flexibility available without delaying the overall project.

The total expected project time $T_e$ is calculated by adding together the expected durations of all activities that lie on the critical path. Because each activity duration involves a degree of

uncertainty, the project variance is obtained by summing the variances of the activities on the critical path. The project standard deviation $\sigma^2(T_e)$ is then derived by taking the square root of this total variance. This approach provides a measure of overall schedule uncertainty and makes it possible to estimate the probability of completing the project within a given deadline.

Despite its usefulness, PERT rests on assumptions that project managers must treat cautiously. First, it assumes precise knowledge of task sequencing within the project diagram, making accuracy in defining dependencies essential. Second, it assumes that resource changes or technical performance affect only task duration. In reality, measures such as overtime may shorten completion time but increase fatigue and reduce quality, creating risks for subsequent tasks. Third, PERT's weighting formula biases results toward the most likely estimate, potentially understating extreme risks (Malcolm et al., 1959).

## 2. Project Crashing

Project crashing is defined as the process of reducing the duration of activities on the critical path from their normal times toward their minimum crash times by allocating additional resources, thereby increasing costs, in order to shorten the overall project completion time (Kelley & Walker, 1959). Project managers often employ project crashing to shorten the project duration without changing scope (AcqNotes, 2023). The principle of crashing lies in the time–cost trade-off curve, which assumes that certain tasks can be accelerated by allocating more resources. While this reduces duration, it increases direct costs, requiring managers to balance earlier completion against higher expenditures (Kelley & Walker, 1959; Babu & Suresh, 1996). Typical crash costs may stem from overtime labor, additional staffing, expedited shipping, or more advanced equipment. The objective is not simply to shorten the project, but to do so cost-effectively by prioritizing critical path activities that yield the greatest time savings per unit of added cost.

Crashing is closely linked to the Critical Path Method (CPM), introduced by Kelley and Walker (1959). CPM is a scheduling technique that identifies the longest sequence of dependent activities and calculates early and late start and finish times, distinguishing critical tasks from those with float. Crashing is then applied selectively to critical activities to compress the overall duration. When uncertainty is significant, CPM is often complemented by PERT, which incorporates probability distributions for task durations. CPM and PERT provide a framework for deterministic and probabilistic schedule analysis, making them suitable for highly uncertain projects, while crashing serves as the decision-making mechanism for time.

To crash a project, the following steps may be performed. First, a targeted finish date is determined. For each task, the project manager should record its crash time, which represents the shortest possible duration, and its crash cost, which is the highest cost incurred when the

task is accelerated. Using this information, the crash slope, or the marginal cost per day saved, can be calculated by applying the formula:

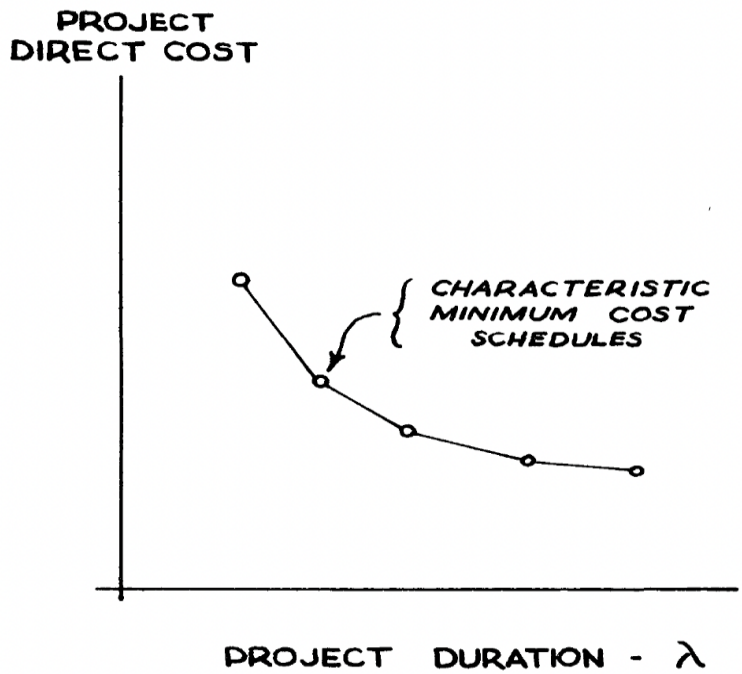
$$Slope = \frac{Crash\ Cost - Normal\ Cost}{Normal\ Time - Crash\ Time} \tag{3}$$

Eq. 3 expresses in dollars per day. Only tasks with a defined slope are considered cashable and can contribute to reducing the overall project duration.

At this stage, it is also useful to visualize the project cost curve. The cost curve, as shown in Figure 2, typically shows the trade-off between project duration and total cost, combining both direct costs (such as overtime, expedited shipping, or extra labor from crashing) and indirect costs (such as overhead or administrative expenses). Initially, as the project is shortened, total costs may decrease because reductions in indirect costs outweigh the increase in direct costs. However, as further crashing is applied, the curve begins to rise since each additional reduction in duration becomes progressively more expensive. The optimal project duration therefore lies at the lowest point on the cost curve, where the balance between crashing costs and indirect cost savings is achieved.

**Figure 2**

*Theoretical Model of Project Crashing and Cost Trade-offs*




*Note.* Figure from *Critical-Path Planning and Scheduling,* by Kelley, J. E., & Walker, M. R., 1959, *Proceedings of the Eastern Joint Computer Conference*. 160–173.

Next, the project manager identifies crash candidates. Only tasks on the current critical path can reduce the project finish time. If there are multiple critical paths, the manager needs to crash at least one task on each path to shorten the project. This crashing step is done in an iterative manner, which means that the crash candidates are considered one at a time and the project finish time is reduced task-by-task. In general, the cheapest task on the critical path is selected first as to minimize the crash cost of the project. If multiple critical paths exist, choose the combination with the lowest total cost that covers all the critical paths. After reducing a chosen task, record the new total duration and update the critical paths of the project. Re-run the previous steps again until the desired finish date is met, or no critical path can be crashed further.

Disadvantages of crashing include increases in direct costs, risks of reduced quality, and team burnout. Crashing is fundamentally expensive, as it requires additional resources for the project. Examples include overtime labor, expedited shipping, hiring temporary staff, and outsourcing. The costs of these extra resources can quickly escalate and strain the project budget if not properly monitored and controlled. Team burnout and quality reduction are closely related. If crashing relies on the same team already assigned to the project, burnout is likely to occur since crashing typically involves extended hours and greater pressure to meet deliverables (Bois et al., 2025; SLM.MBA, 2024). Prolonged fatigue and reduced morale can compromise attention to detail, increase error rates, and ultimately diminish the overall quality of the final output.

## 3. Applications of PERT and Project Crashing

PERT and project crashing address two common project challenges: uncertainty and time pressure. PERT improves planning by using probabilistic estimates to forecast schedules and risks, while crashing shortens project duration by adding resources to critical tasks. Together, they allow projects to be planned realistically and adjusted when deadlines are at risk.

### 3.1. Application of PERT

The Program Evaluation and Review Technique (PERT) was first applied in 1958 to the U.S. Navy's Polaris Fleet Ballistic Missile program, one of the most complex research and development efforts of its time. Traditional scheduling methods were inadequate for managing thousands of interdependent activities under high uncertainty, so PERT was developed to provide a structured, probabilistic approach to planning and control.

In practice, PERT modeled the program as a network of events and activities, with time estimates collected from technical experts. This allowed analysts to identify the critical path, determine scheduling flexibility through slack, and assess the probability of meeting

established deadlines. The system was first tested on subsystems such as propulsion before being scaled up to cover the full missile program. Using high-speed computing, the Navy was able to process networks of thousands of activities, update forecasts, and simulate the effects of proposed changes on overall completion.

The practical benefits were substantial. PERT enabled early detection of schedule risks, supported management-by-exception by highlighting critical tasks, and improved coordination among contractors and government agencies. Within 17 months of implementation, PERT covered over 2,000 events and 3,000 activities, and its use in simulation runs significantly reduced the likelihood of missing critical deadlines. The Polaris case demonstrated PERT's effectiveness as a quantitative decision-support tool for managing large, uncertain, and technologically advanced projects.

### 3.2. Application of Project Crashing

The 2004 Athens Olympic Games are often cited as a prominent case of project crashing under extreme time pressure. In the years before the Games, Greece struggled to complete the extensive infrastructure needed, including stadiums, the Olympic village, transport networks, and telecommunications. Delays stemmed from slow decision-making, complex approvals, and difficulties coordinating contractors. By 2002, international observers expressed concern that key venues would not be ready, creating risks for both national reputation and the successful execution of the event.

To address these challenges, managers employed schedule compression consistent with crashing. Critical path activities were accelerated by adding labor, running construction shifts around the clock, and bringing in additional contractors to execute tasks in parallel. Procurement was expedited, and some materials were obtained at higher costs to avoid delays. These measures sharply increased expenditure but were considered unavoidable given the immovable Olympic deadline.

The results were mixed. Essential venues were completed on time and prevented a logistical crisis, but the financial impact was severe. Flyvbjerg and Stewart (2012) report that Athens became the most expensive Olympics in history at that point, with costs surpassing €9 billion, much of it linked to late-stage acceleration. The compressed schedule also strained quality control and raised concerns about long-term sustainability. The case illustrates the critical role of crashing on large projects with fixed deadlines. It shows that managers may accept higher costs and risks to ensure timely delivery, while recognizing that crashing is a short-term remedy than a sustainable strategy.

# C. ANALYSIS

The analysis of the SkyTrack project is structured in three stages. First, the baseline plan is outlined to identify the task structure, dependencies, critical path, and estimated duration. Second, the baseline plan is analyzed to know the probability of completing the project within 90 days and explores the optimum decision between time and costs. And finally, the baseline plan is tested against potential risk scenarios to assess its impact on project schedule.

## 1. SkyTrack Baseline Plan

Based on the initial activities and duration identified by SkyTrack team (see Appendix 1), the task was organized into a Work Breakdown Structure (WBS) based on the project's objective and key deliverables (see Figure 2). The expected duration of each task was also calculated using the optimistic, most-likely, and pessimistic duration estimates from the initial plan, along with its associated variance of each task ($T_e$ = expected duration, var = deviation of expected duration, see Table 1).

**Figure 2**
*Work Breakdown Structure (WBS) of SkyTrack Project*

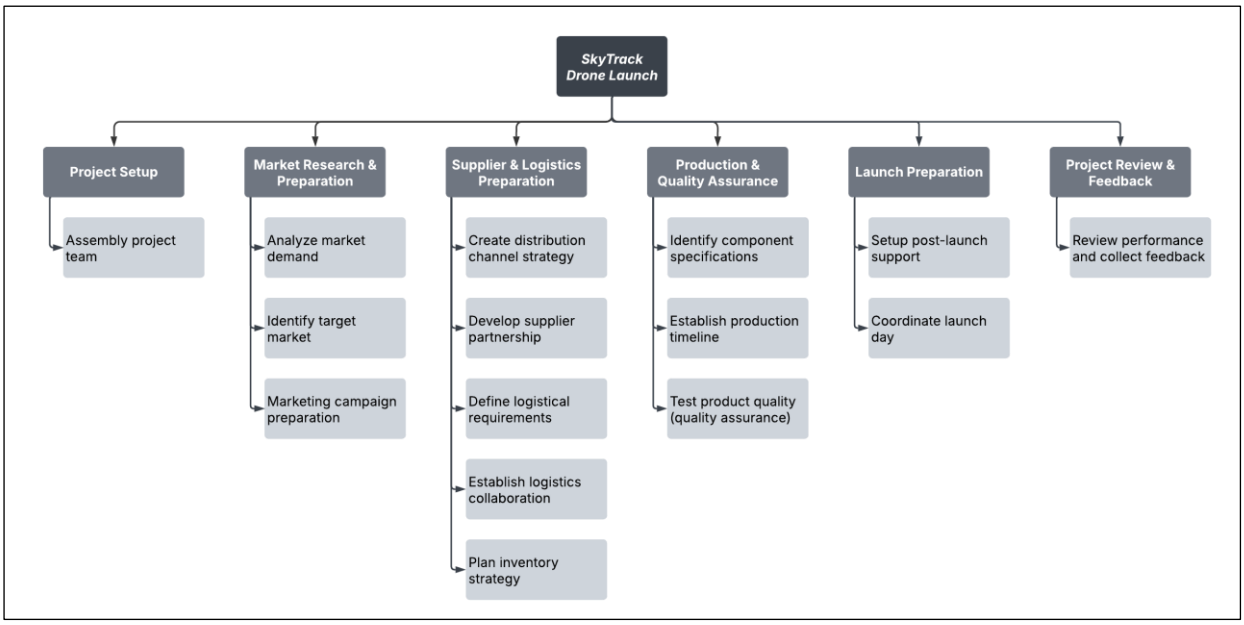


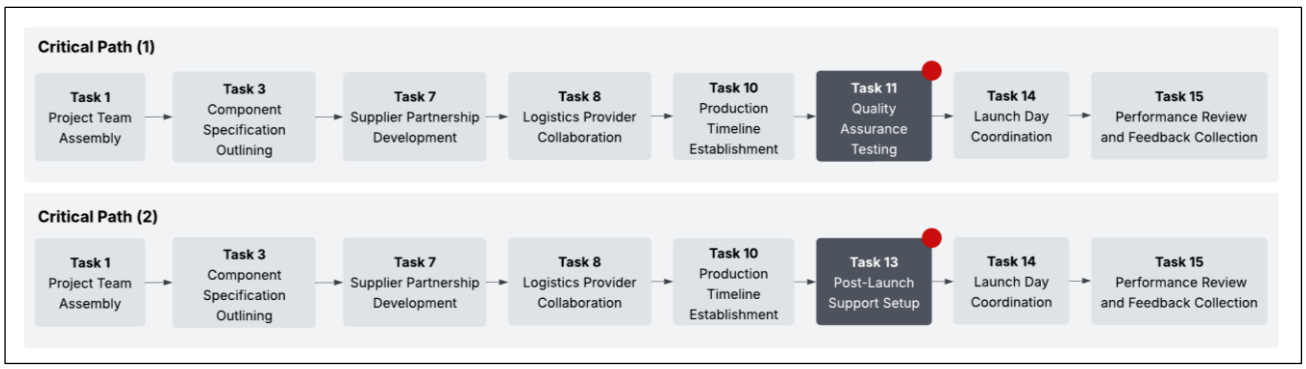
**Table 1**
*Activity Time Estimates and Dependencies*

| Task ID | Task | $T_e$ | var | Predecessor | | | |
|---|---|---|---|---|---|---|---|
| 1 | Project team assembly | 6 | 2.78 | | | | |
| 2 | Define logistical requirements | 11 | 4.00 | (1) | | | |
| 3 | Component specification outlining | 16 | 5.44 | (1) | | | |
| 4 | Market demand analysis | 21 | 5.44 | (1) | | | |
| 5 | Target market identification | 11 | 1.78 | (4) | | | |
| 6 | Distribution channel strategy | 16 | 4.00 | (5) | | | |
| 7 | Supplier partnership development | 21 | 4.00 | (2) | (3) | | |
| 8 | Logistics provider collaboration | 16 | 5.44 | (7) | | | |
| 9 | Inventory management planning | 11 | 2.78 | (7) | | | |
| 10 | Production timeline establishment | 11 | 1.78 | (8) | (9) | | |
| 11 | Quality assurance testing | 16 | 4.00 | (10) | | | |
| 12 | Marketing campaign preparation | 16 | 2.78 | (5) | (7) | | |
| 13 | Post-launch support setup | 16 | 1.78 | (11SS) | | | |
| 14 | Launch day coordination | 6 | 1.78 | (6) | (11) | (12) | (13) |
| 15 | Performance review and feedback | 11 | 4.00 | (14) | | | |

**Critical Path Analysis**

Based on the expected duration and its varience, we identify SkyTrack project critical path using the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM). Two critical paths were identified (see Figure 2), with the overall project estimated to take **103 days** to be completed. Tasks on these paths have zero slack, meaning that any delay in their execution will directly impact on the project's final completion date.

**Figure 2**
*Identified critical path of SkyTrack project*

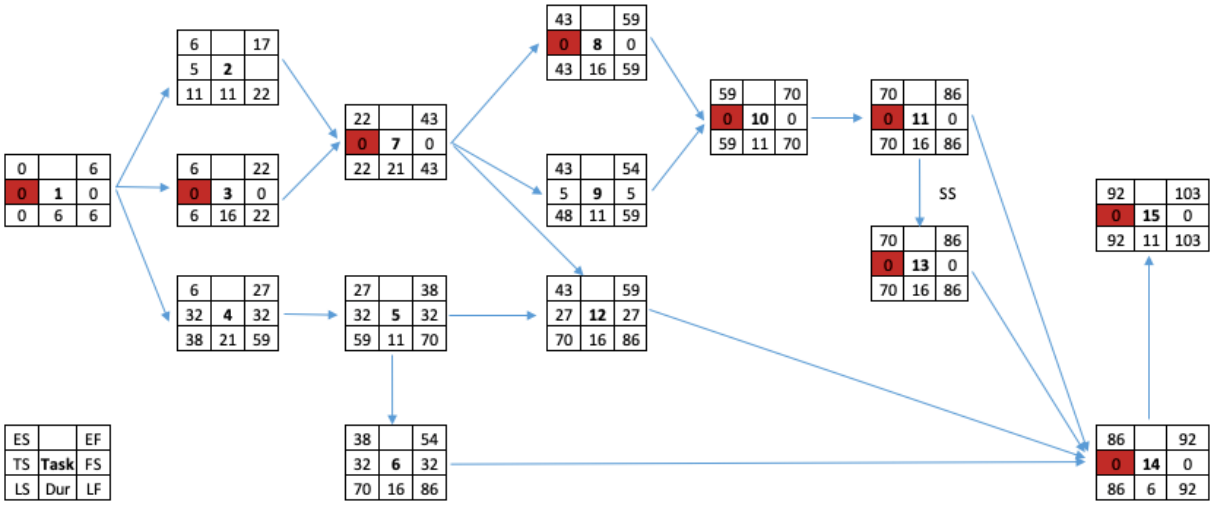



This dual critical path structure emerges from a Start-to-Start (SS) relationship between Task 11 (Quality Assurance Testing) and Task 13 (Post-Launch Support Setup) (see network diagram in Figure 3 to see dependencies of each task). Both tasks begin after Task 10

(Establish Production Timeline) but proceed independently until Task 14 (Launch Day). This structure introduces additional complexity, as both paths must be monitored closely to prevent schedule delays.

**Figure 3**

*Network diagram of SkyTrack initial project plan*




Building on this initial plan, the following analysis evaluates the SkyTrack project under different scenarios using PERT and crashing techniques. The objective is to examine how time, cost, and risk interact, and to highlight the trade-offs managers must consider when faced with tight deadlines or external pressures.

## 2. Baseline Plan Analysis

Before considering alternative scenarios, it is essential to first establish the baseline plan as a reference point. The baseline provides the expected project duration and cost under normal conditions, while also highlighting the gap between the current schedule and the 90-day target. Understanding this starting position allows AeroDynamics to better evaluate how strategies such as crashing, task adjustments, or supplier changes can shift the balance between time, cost, and risk.

### 2.1. Probability of Completion (PERT Analysis)

Following the PERT methodology, which assumes project durations follow a normal distribution, we estimated the probability of completing the project within **90 days**, significantly shorter than the **103-day** expectation.

Two critical paths were analyzed independently due to the Start-to-Start dependency structure (Tasks 11 and 13). The calculations for each path are summarized below:

**Table 3**
*SkyTrack's Priority Matrix*

| *Metric* | **Path 1** | **Path 2** |
|---|---|---|
| *Expected Duration* | 103 days | 103 days |
| *Standard Deviation (σ)* | 5.41 | 5.20 |
| *Scheduled Duration (T)* | 90 days | 90 days |
| *Probability of On-Time Completion* | **0.81%** | **0.62%** |

These results indicate that the likelihood of completing the project within 90 days is less than 1% for both paths. Therefore, a 90-day schedule would be considered highly aggressive and risky, with a strong likelihood of delay unless project scope is reduced, resources are added, or task durations are shortened.

The probability estimates provide valuable insights for AeroDynamics Inc. on planning and decision-making, particularly in evaluating schedule feasibility and identifying areas requiring contingency planning.

## 2.2. Case Analysis of Project Initial Plan

This section evaluates how the initial plan performs against the 90-day target. By applying PERT and cost – time analysis, we test whether the plan can realistically meet the deadline and what trade-offs appear when different crashing options are used.

**90-Days Completion Time**

The project has an expected duration of 103 days with a probability of on-time completion in 90 days of <1%. To raise this to 95% confidence, the expected duration must be reduced to approximately 81 days. This requires shortening 22 days along the critical path, which can be accomplished by crashing.

Using the crashing options and costs shown in the Appendix 2, the required reductions occur along the critical path in this sequence, based on the crashing time available and the cheap crashing cost on priority.

Task1 (×2d), Task3 (×5d), Task10 (×5d), Task7 (×5d), Task14 (×1d), Task15(×2d), Task8 (×2d)

The cumulative crash cost is $18,500, increasing direct costs from $55,500 to $74,000. Indirect costs decline from $80,000 to $60,200 (−$900/day × 22 days). The resulting total cost is $134,200, which is $1,300 lower than initial plan due to the reduction in indirect cost.

If 95% confidence by Day 90 is mandatory, adopt the above crashing plan exactly as sequenced. Tighten governance on all tasks on the critical path post-crash, and pre-approve overtime/parallel work to avoid re-expanding the path.

**Optimum Cost-Time Point**

Analysis of the cost-time trade-off curve shows that the lowest total cost, approximately $131,200, is achieved at a project duration of 91 days. Compared with the initial schedule of 103 days at $135,500, this represents a cost saving of about $4,300. At this point, indirect costs fall sharply due to the shorter duration, while direct costs remain controlled without the steep increases associated with excessive crashing.

These findings highlight that the most cost-efficient plan is not necessarily the shortest one. For the SkyTrack project, a 91-day schedule offers the optimal balance between time and cost, keeping overall expenditure under control while avoiding the risks, inefficiencies, and resource strain associated with overly aggressive acceleration.

**Minimum Expected Completion Time and Cost**

If the project is continuously crash to its limit, the shortest achievable duration is **79 days**. To reach this point, several high-cost crashing options must be applied, significantly increasing direct expenditure.

At 79 days, the project's cost breakdown is as follows:

- **Direct cost:** $80,500
- **Indirect cost:** $58,400
- **Total cost: $138,900**

Compared with the initial schedule of **103 days and $135,500**, this minimum-time scenario shortens the project by **24 days** but increases the overall cost by **$3,400**. This outcome demonstrates diminishing returns: while the project duration is reduced, the higher total cost and increased resource pressure make this option unattractive except in cases where time-to-market is the overriding strategic priority.

**Table 3**

*Summarizes the results of crashing analysis*

| Scenario | Duration (days) | Time of crashing | Total Cost | Likelihood of completion in 90 days |
|---|---|---|---|---|
| Reference case | 103 | N/A | $135,500 | 0.62% |
| 95% likelihood of finishing in 90 days (aggressive crash) | 81 | 22 | $134,200 | 95% |
| Optimum cost–time point | 91 | 12 | $131,200 | 42% |
| Shortest possible duration | 79 | 24 | $138,900 | 98% |

The summary table below highlights clear trade-offs across the four scenarios. The initial plan (103 days, $135,500) serves as a benchmark but has less than a 1% chance of meeting the board's 90-day target, making it unrealistic without intervention.

Scenario 1 (81 days, $134,200) achieves a 95% likelihood of finishing in 90 days, but only through aggressive crashing of 22 days' worth of work. While this option slightly reduces cost compared with the baseline (a saving of about $1,300 or 1%), it introduces high resource strain and execution risk. Nonetheless, it is still more efficient than maintaining the current baseline.

Scenario 2 (91 days, $131,200) represents the optimum cost–time balance, with the lowest total cost, saving about $4,300 (3%) compared with the initial plan while also reducing the schedule by nearly two weeks. At this point, the probability of completing within 90 days rises to 42%, which, while not certain, offers a pragmatic middle ground between speed and cost efficiency.

Scenario 3 (79 days, $138,900) delivers the fastest possible completion and a near-guaranteed 98% chance of meeting the 90-day target, but it does so at the highest cost. At $3,400 (2.5%) more than baseline, and $7,700 more expensive than Scenario 2, this option represents poor value unless time-to-market is the overriding strategic priority.

## 3. What-If Scenario Analysis

The analysis below explores how changes in assumptions, such as supplier delays or faster task completion, will affect the project. The aim is to see how these shifts influence the cost and time balance and its relevancy for SkyTracks 's ability to deliver on schedule.

### 3.1. Scenario 4: Impact of Overseas Supplier Partnership

In this scenario, the **Supplier Partnership Development task (Task 7)** was extended from 21 to 30 days, assumed at no additional cost. This adjustment pushed the project duration from **103 to 112 days**. The critical paths remain unchanged, but each becomes longer.

After applying crashing, only where the crash cost per day stayed below the $900 indirect cost saving, the cheapest plan remained at $131,200. The key difference is that the optimum point reached at 100 days instead of 91 days. The shortest achievable schedule after full crashing is also extended to 88 days instead of the original 79 days. By that even with the all the effort of crashing to make the expected duration by 88 days, the likelihood of SkyTrack launched on time is only 64%.

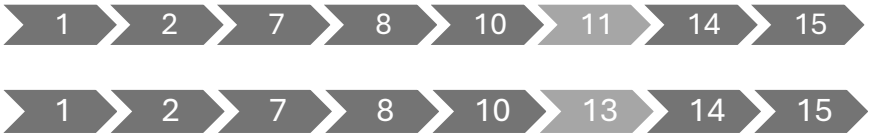
**Table 4**
*Initial Plan and Scenario 4 comparison*

| Metric | Initial Plan (Task 7 = 21 days) | Scenario 4 (Task 7 = 30 days) |
|---|---|---|
| Expected Duration | 103 days | 112 days |
| Optimum Duration | 91 days | 100 days |
| Optimum Total Cost | $131,200 | $131,200 |
| Shortest Duration | 79 days | 88 days |
| Shortest Total Cost | $138,900 | $138,900 |

This result shows that supplier delays do not always translate into higher costs, but they do consume valuable time, and in many cases, time is money. Even if the budget remains the same, losing nine days of efficiency makes the schedule tighter. For SkyTrack, this highlights the importance of securing supplier timelines early and maintaining contingency plans, since once time is lost to delays, even the most effective crashing strategy cannot recover it. Better have a plan on mitigating the risk and closely government critical task.

### 3.2. Scenario 5: Impact of Reduced Task Duration

In this scenario, the expected duration of **Task 3 (Component Specification Outlining)** is reduced by seven days at no additional cost. This adjustment lowers the project's duration from 103 days to 98 days, while the total cost remains unchanged at $131,500. The critical path shifts slightly, now including Task 2 instead of Task 3, resulting in two new parallel critical paths:




The crashing analysis shows that the optimum cost-time point occurs at 89 days with a total cost of $128,900. At this optimal point, the project has nine days shorter and save $2,600. Furthermore, this scenario gave two days faster and $2,300 cheaper than the previous optimum identified in Scenario 2 (91 days, $131,200). However, with 89 days expected duration, the project only has 57% of completion on time.

The shortest achievable schedule under this scenario is 77 days at a total cost of $136,600, bring us 99% of confident that we can get in the market in 90 days. However, this heavily compressed schedule requires extensive crashing of critical activities and should only be considered if time-to-market urgency outweighs cost efficiency.

# 4. CONCLUSION AND RECOMMENDATION

The SkyTrack project was designed with an ambitious 90-day launch target, reflecting AeroDynamics Inc.'s priority on speed to market as a critical success factor. However, the PERT/CPM analysis showed that the baseline schedule requires 103 days, with less than a 1% probability of meeting the board's target. This confirms that the original plan, while well structured, cannot meet expectations without intervention.

To achieve 95% confidence of finishing within 90 days, the expected duration must be reduced to 81 days, which requires crashing seven tasks for a total of 22 days. While this option is feasible, it introduces heavy resource strain and results in a 2.3% cost increase, trading efficiency for certainty. If market conditions demand an even earlier launch, all possible crashing can be applied, reducing the schedule to 79 days. In this case, the project achieves 95% confidence of finishing in 88 days and 98% likelihood of finishing within 90 days, though at an extra 2.5% cost compared to the baseline. Such an approach would require continuous contingency planning and close governance of all critical tasks.

The supplier delay scenario (Scenario 4) demonstrated the project's high vulnerability to external risks. A delay in a single critical activity, such as supplier partnership development, extends the overall schedule by the same amount, reducing the chance of meeting the 90-day target even under full crashing (to around 57%). This highlights the importance of locking in supplier timelines early and developing contingency plans. By contrast, the reduced task duration scenario (Scenario 5) showed that shortening one critical activity by seven days lowered the baseline to 98 days and shifted the optimum cost–time point to 89 days at $128,900. This improves cost efficiency and increases the likelihood of meeting the 90-day target. However, once a task reaches its minimum, no further compression is possible, and shortening one task may shift the critical path to another sequence. This underscores the need to continuously monitor the critical path, as even slight changes can alter which activities control project completion.

Overall, the scenario analysis highlights that achieving AeroDynamics' 90-day target depends on carefully balancing time, cost, and risk, while keeping Rebecca Sinclair's three launch objectives in focus: resilient supply chains, effective inventory management, and reliable logistics. If tighter deadlines are imposed, the project must accelerate critical tasks, supported by inventory buffers and logistics capacity. If cost efficiency is prioritised, the schedule should stabilise around the optimum plan, reinforced by resilient supplier agreements to absorb delays. If supplier delays occur, contingency sourcing and early contracting are essential. Where early task efficiencies are available, they should be captured,

but managers must monitor for shifts in the critical path. In all cases, success will rely on proactive supplier governance, disciplined inventory planning, and robust logistic systems to flex with changing board or market demands, ensuring SkyTrack's timely launch.

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# APPENDICES

**Appendix 1**

*Skytrack Project task and durations*

| Task | Duration (days) | | |
|---|---|---|---|
| | **Optimistic** | **Most Likely** | **Pessimistic** |
| Project team assembly | 3 | 5 | 13 |
| Define logistical requirements | 7 | 10 | 19 |
| Component specification outlining | 11 | 15 | 25 |
| Market demand analysis | 16 | 20 | 30 |
| Target market identification | 9 | 10 | 17 |
| Distribution channel strategy | 12 | 15 | 24 |
| Supplier partnership development | 17 | 20 | 29 |
| Logistics provider collaboration | 11 | 15 | 25 |
| Inventory management planning | 8 | 10 | 18 |
| Production timeline establishment | 9 | 10 | 17 |
| Quality assurance testing | 12 | 15 | 24 |
| Marketing campaign preparation | 13 | 15 | 23 |
| Post-launch support setup | 14 | 15 | 22 |
| Launch day coordination | 4 | 5 | 12 |
| Performance review and feedback | 7 | 10 | 19 |

**Appendix 2**

*Skytrack Project task and durations crashing for Scenario 1,2 and 3*

| Project duration | Crashed activities | Crash cost | Direct cost | Indirect cost | Total cost | Notice |
|---|---|---|---|---|---|---|
| 103 | | | $55,500 | $80,000 | $135,500 | |
| 102 | 1 | $250 | $55,750 | $79,100 | $134,850 | |
| 101 | 1 | $250 | $56,000 | $78,200 | $134,200 | |
| 100 | 3 | $400 | $56,400 | $77,300 | $133,700 | |
| 99 | 3 | $400 | $56,800 | $76,400 | $133,200 | |
| 98 | 3 | $400 | $57,200 | $75,500 | $132,700 | |
| 97 | 3 | $400 | $57,600 | $74,600 | $132,200 | |
| 96 | 3 | $400 | $58,000 | $73,700 | $131,700 | |
| 95 | 10 | $800 | $58,800 | $72,800 | $131,600 | |
| 94 | 10 | $800 | $59,600 | $71,900 | $131,500 | |
| 93 | 10 | $800 | $60,400 | $71,000 | $131,400 | |
| 92 | 10 | $800 | $61,200 | $70,100 | $131,300 | |
| **91** | **10** | **$800** | **$62,000** | **$69,200** | **$131,200** | **Scenario 2** |
| 90 | 7 | $1,00›0 | $63,000 | $68,300 | $131,300 | |
| 89 | 7 | $1,000 | $64,000 | $67,400 | $131,400 | |
| 88 | 7 | $1,000 | $65,000 | $66,500 | $131,500 | |
| 87 | 7 | $1,000 | $66,000 | $65,600 | $131,600 | |
| 86 | 7 | $1,000 | $67,000 | $64,700 | $131,700 | |
| 85 | 14 | $1,000 | $68,000 | $63,800 | $131,800 | |
| 84 | 15 | $1,000 | $69,000 | $62,900 | $131,900 | |
| 83 | 15 | $1,000 | $70,000 | $62,000 | $132,000 | |
| 82 | 8 | $2,000 | $72,000 | $61,100 | $133,100 | |
| **81** | **8** | **$2,000** | **$74,000** | **$60,200** | **$134,200** | **Scenario 1** |
| 80 | 11, 13 | $3,250 | $77,250 | $59,300 | $136,550 | |
| **79** | **11, 13** | **$3,250** | **$80,500** | **$58,400** | **$138,900** | **Scenario 3** |

**Appendix 3**

*Skytrack Project task and durations crashing for Scenario 4*

| Project duration | Crashed activities | Crash cost | Direct cost | Indirect cost | Total cost | Notice |
|---|---|---|---|---|---|---|
| 112 | | | $55,500 | $80,000 | $135,500 | |
| 111 | 1 | $250 | $55,750 | $79,100 | $134,850 | |
| 110 | 1 | $250 | $56,000 | $78,200 | $134,200 | |
| 109 | 3 | $400 | $56,400 | $77,300 | $133,700 | |
| 108 | 3 | $400 | $56,800 | $76,400 | $133,200 | |
| 107 | 3 | $400 | $57,200 | $75,500 | $132,700 | |
| 106 | 3 | $400 | $57,600 | $74,600 | $132,200 | |
| 105 | 3 | $400 | $58,000 | $73,700 | $131,700 | |
| 104 | 10 | $800 | $58,800 | $72,800 | $131,600 | |
| 103 | 10 | $800 | $59,600 | $71,900 | $131,500 | |
| 102 | 10 | $800 | $60,400 | $71,000 | $131,400 | |
| 101 | 10 | $800 | $61,200 | $70,100 | $131,300 | |
| **100** | **10** | **$800** | **$62,000** | **$69,200** | **$131,200** | **Lowest cost** |
| 99 | 7 | $1,000 | $63,000 | $68,300 | $131,300 | |
| 98 | 7 | $1,000 | $64,000 | $67,400 | $131,400 | |
| 97 | 7 | $1,000 | $65,000 | $66,500 | $131,500 | |
| 96 | 7 | $1,000 | $66,000 | $65,600 | $131,600 | |
| 95 | 7 | $1,000 | $67,000 | $64,700 | $131,700 | |
| 94 | 14 | $1,000 | $68,000 | $63,800 | $131,800 | |
| 93 | 15 | $1,000 | $69,000 | $62,900 | $131,900 | |
| 92 | 15 | $1,000 | $70,000 | $62,000 | $132,000 | |
| 91 | 8 | $2,000 | $72,000 | $61,100 | $133,100 | |
| 90 | 8 | $2,000 | $74,000 | $60,200 | $134,200 | |
| 89 | 11, 13 | $3,250 | $77,250 | $59,300 | $136,550 | |
| **88** | **11, 13** | **$3,250** | **$80,500** | **$58,400** | **$138,900** | **Shortest time** |

**Appendix 4**

*Skytrack Project task and durations crashing for Scenario 5*

| Project duration | Crashed activities | Crash cost | Direct cost | Indirect cost | Total cost | Notice |
|---|---|---|---|---|---|---|
| 98 | | | $51,500 | $80,000 | $131,500 | |
| 97 | 1 | $250 | $51,750 | $79,100 | $130,850 | |
| 96 | 1 | $250 | $52,000 | $78,200 | $130,200 | |
| 95 | 2 | $500 | $52,500 | $77,300 | $129,800 | |
| 94 | 2 | $500 | $53,000 | $76,400 | $129,400 | |
| 93 | 10 | $800 | $53,800 | $75,500 | $129,300 | |
| 92 | 10 | $800 | $54,600 | $74,600 | $129,200 | |
| 91 | 10 | $800 | $55,400 | $73,700 | $129,100 | |
| 90 | 10 | $800 | $56,200 | $72,800 | $129,000 | |
| **89** | **10** | **$800** | **$57,000** | **$71,900** | **$128,900** | **Lowest Cost** |
| 88 | 7 | $1,000 | $58,000 | $71,000 | $129,000 | |
| 87 | 7 | $1,000 | $59,000 | $70,100 | $129,100 | |
| 86 | 7 | $1,000 | $60,000 | $69,200 | $129,200 | |
| 85 | 7 | $1,000 | $61,000 | $68,300 | $129,300 | |
| 84 | 7 | $1,000 | $62,000 | $67,400 | $129,400 | |
| 83 | 14 | $1,000 | $63,000 | $66,500 | $129,500 | |
| 82 | 15 | $1,000 | $64,000 | $65,600 | $129,600 | |
| 81 | 15 | $1,000 | $65,000 | $64,700 | $129,700 | |
| 80 | 8 | $2,000 | $67,000 | $63,800 | $130,800 | |
| 79 | 8 | $2,000 | $69,000 | $62,900 | $131,900 | |
| 78 | 11,13 | $3,250 | $72,250 | $62,000 | $134,250 | |
| **77** | **11,13** | **$3,250** | **$75,500** | **$61,100** | **$136,600** | **Shortest Time** |